

# **BIO00056I Summary of main topics & core concepts**

**Evolutionary Genetics & Population Genomics**

Ferrari, Dasmahapatra, Harper, Jeffares

2026-09-22

## **Table of contents**

|           |                                                                       |           |
|-----------|-----------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Deterministic vs. Stochastic Evolutionary Processes</b>            | <b>2</b>  |
| <b>2</b>  | <b>Molecular Mechanisms of Mutation and Substitution</b>              | <b>2</b>  |
| <b>3</b>  | <b>Genetic Drift and Effective Population Size (<math>N_e</math>)</b> | <b>3</b>  |
| <b>4</b>  | <b>Natural Selection: Models, Dynamics, and Modes</b>                 | <b>3</b>  |
| <b>5</b>  | <b>Mechanisms Maintaining Genetic Variation</b>                       | <b>4</b>  |
| <b>6</b>  | <b>Population Structure and Subgroup Differentiation</b>              | <b>5</b>  |
| <b>7</b>  | <b>Neutral Theory and the Molecular Clock</b>                         | <b>6</b>  |
| <b>8</b>  | <b>Phylogenetic Analysis and Evolutionary Trees</b>                   | <b>7</b>  |
| <b>9</b>  | <b>Conservation Genetics and Molecular Forensics</b>                  | <b>7</b>  |
| <b>10</b> | <b>Integrative Real-World Applications &amp; Quantitative Methods</b> | <b>8</b>  |
| <b>11</b> | <b>Population Genomics and Genome-Wide Selection Dynamics</b>         | <b>8</b>  |
|           | 11.0.1 Genomic Patterns Across Modes of Selection . . . . .           | 9         |
| <b>12</b> | <b>Comparative Genomics, Evolutionary Rates, and Divergence</b>       | <b>10</b> |
| <b>13</b> | <b>Continuous Variation, Phenotypic Components, and Heritability</b>  | <b>11</b> |
| <b>14</b> | <b>Quantitative Trait Loci (QTL) Mapping and GWAS</b>                 | <b>12</b> |

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| <b>15 Genome Evolution and Structural Dynamics</b>                   | <b>12</b> |
| 15.0.1 Structural Organization: Prokaryotes vs. Eukaryotes . . . . . | 12        |

## 1 Deterministic vs. Stochastic Evolutionary Processes

- **Definition of Evolution:** Fundamental change in allele or genotype frequencies in a population from one generation to the next.
- **Four Primary Mechanisms:** Allele frequency changes are driven by natural selection, genetic drift, mutation, and gene flow.
- **Natural Selection (Deterministic):** Acts non-randomly on fitness differences between phenotypes, requiring phenotypic variation, a link between phenotype and fitness, and a genetic basis (heritability). Drives adaptation and typically reduces genetic variation.
- **Mutation (Stochastic):** The ultimate random source of all new genetic variation, occurring at very low frequencies.
- **Genetic Drift (Stochastic):** Random fluctuations in allele frequencies between generations due to sampling error, leading to a loss of genetic diversity—particularly in small populations.
- **Gene Flow / Migration:** Spatial movement between populations that reduces genetic divergence between subpopulations and restores variability.
- **Null Models:** Hardy-Weinberg equilibrium provides the baseline null model to predict expected genotype frequencies when no evolutionary forces act.

---

## 2 Molecular Mechanisms of Mutation and Substitution

- **Mutation vs. Substitution:** A mutation is any newly arising genetic change in an individual; a substitution is a mutation that has successfully spread to fixation across the population.
- **Point Mutations:** Base-pair substitutions in coding regions are classified as synonymous (silent) or non-synonymous (amino acid altering).
- **Indels & Reading Frames:** Insertions and deletions (indels) within coding sequence regions can alter the reading frame via frameshift mutations.
- **Structural & Chromosomal Alterations:** Gene duplications create multigene families and pseudogenes (e.g., human  $\beta$ -globin gene family). Larger structural mutations include transposons, duplications, rearrangements, inversions, translocations, fission, fusion, and ploidy changes.
- **Mutational Fates:** The fate of a new mutation (fixation, loss, or balancing) is governed by selection and drift; most new neutral or deleterious mutations are rapidly lost to drift or purifying selection.

---

### 3 Genetic Drift and Effective Population Size ( $N_e$ )

- **Fixation Rules:** For a neutral allele, the probability of eventual fixation equals its current initial allele frequency ( $p$ ), and the time to fixation is directly proportional to population size.
- **Concept of  $N_e$ :** Effective population size ( $N_e$ ) represents the size of an idealized population that experiences the same rate of inbreeding or loss of genetic diversity as the real population.
- **$N_e$  vs. Census Size ( $N$ ):**  $N_e$  is almost universally smaller than  $N$  due to real-world deviations from ideal assumptions.
- **Factors Reducing  $N_e$ :**
  - **Reproductive Skew:** Unequal numbers of breeding females ( $N_f$ ) and males ( $N_m$ ) decrease  $N_e$  via:

$$N_e = \frac{4N_f N_m}{N_f + N_m}$$

- **Variance in Family Size:** High variance in individual reproductive output reduces  $N_e$  relative to  $N$ .
- **Population Fluctuations & Bottlenecks:**  $N_e$  across generations is dictated by the harmonic mean:

$$\frac{1}{N_e} = \frac{1}{t} \sum \frac{1}{N_i}$$

Bottlenecks dramatically reduce  $N_e$  and amplify genetic drift.

---

### 4 Natural Selection: Models, Dynamics, and Modes

- **Fitness Measures:**
  - **Absolute Fitness ( $W$ ):** Genotype-specific rate of increase predicting absolute population numbers.
  - **Relative Fitness ( $w$ ):** Fitness relative to the fittest genotype ( $w_{\max} = 1$ ).

- **Selection Coefficient ( $s$ ):** Measure of the strength of selection against a genotype ( $w = 1 - s$ ).
- **General Selection Model:** The per-generation change in allele frequency ( $\Delta q$ ) depends on initial frequencies ( $p, q$ ) and relative fitness values ( $w_1, w_2, w_3$ ):

$$\Delta q = \frac{pq[q(w_3 - w_2) - p(w_1 - w_2)]}{\bar{w}}$$

where population mean fitness is:

$$\bar{w} = p^2 w_1 + 2pq w_2 + q^2 w_3$$

- **Dominance Effects on Selection Dynamics:** Selection against a dominant allele is rapid initially; selection against a recessive allele slows down substantially as the allele becomes rare because it is “hidden” from selection in heterozygous carriers.
- **Modes of Selection:**
  - **Directional Selection:** Favors one extreme phenotype, eliminating deleterious alleles over time without reaching a stable polymorphic equilibrium.
  - **Stabilizing Selection:** Favors intermediate phenotypes, reducing phenotypic variance.
  - **Disruptive Selection:** Favors extreme phenotypes at both ends of the spectrum over intermediate ones.
- **Frequency-Dependent Selection (FDS):** Occurs when genotype fitness depends on its frequency in the population. Negative FDS maintains polymorphism and leads to a stable equilibrium.

---

## 5 Mechanisms Maintaining Genetic Variation

- **Mutation-Selection Balance:** Rare, severely disadvantageous alleles persist at an equilibrium frequency where their rate of removal by selection equals their rate of creation by recurrent mutation ( $\mu$ ).
  - **Dominant Deleterious Alleles:**  $q \approx \frac{\mu}{s}$
  - **Recessive Deleterious Alleles:**  $q \approx \sqrt{\frac{\mu}{s}}$
- **Heterozygote Advantage (Overdominance):** Occurs when heterozygotes have higher fitness than either homozygote ( $w_{AA} = 1 - s$ ,  $w_{Aa} = 1$ ,  $w_{aa} = 1 - t$ ), leading to a stable equilibrium  $q = \frac{s}{s+t}$ .

- **Segregational Load:** The persistent reduction in average population fitness below 1 due to continuous Mendelian emergence of lower-fitness homozygotes.
  - **Genotype  $\times$  Environment ( $G \times E$ ) Interactions:** Spatial or temporal environmental heterogeneity where relative fitness rankings reverse across environments, preserving variation.
  - **Genetic Correlations & Trade-offs:** Driven by pleiotropy or linkage disequilibrium, creating evolutionary trade-offs and co-adapted trait complexes.
- 

## 6 Population Structure and Subgroup Differentiation

- **Panmixia vs. Population Structure:** Panmixia represents random mating with no allele frequency differences between groups. Reduced gene flow creates population structure, leading to divergence in allele frequencies between subpopulations.
- **Evolutionary Drivers:**
  - **Gene Flow:** Reduces genetic differentiation between populations.
  - **Genetic Drift:** Increases differentiation between isolated populations.
  - **Selection:** Increases differentiation under divergent environmental pressures, but decreases differentiation under universally adaptive selection.
- **Fixation Index ( $F_{ST}$ ):** Measures genetic differentiation due to population subdivision, which reduces expected heterozygosity relative to total population expectation (Wahlund effect).

$$F_{ST} = \frac{H_T - H_S}{H_T}$$

- $H_S$ : Average expected heterozygosity across subpopulations.
- $H_T$ : Total expected heterozygosity calculated from mean allele frequencies across all subpopulations.
- $F_{ST} = 0$ : No allele frequency differences between populations (panmixia).
- $F_{ST} = 1$ : Alternate alleles fixed in different populations.
- **Isolation by Distance (IBD):** Pattern where genetic distance ( $F_{ST}$ ) increases continuously with geographic distance (e.g., Florida black bears).
- **Locus-Specific Selection Signals:** Loci under divergent selection show significantly elevated  $F_{ST}$  compared to neutral background genome regions (e.g., wing pattern loci B/D and Yb in *Heliconius melpomene*).
- **Clustering & Multidimensional Methods:**
  - **Principal Component Analysis (PCA):** Non-tree method to simplify and visualize high-dimensional genomic variation and genetic distance.

- **STRUCTURE Analysis:** Model-based Bayesian clustering method assigning individuals to population clusters to detect migration and admixture across physical barriers.
- 

## 7 Neutral Theory and the Molecular Clock

- **Neutral Theory of Molecular Evolution:** Formulated by Motoo Kimura (1983), stating that most genetic variation within and between species is selectively neutral (having little or no effect on fitness).
- **Mathematical Derivation of the Molecular Clock:**
  - In a diploid population of size  $N$ , with a neutral mutation rate  $\mu$  per gamete per generation, there are  $2N\mu$  new mutations each generation.
  - The fixation probability  $u$  of any single neutral mutation by random genetic drift equals its initial frequency:

$$u = \frac{1}{2N}$$

- The substitution rate  $k$  (number of fixed substitutions per unit time) is:

$$k = 2N\mu \cdot u = 2N\mu \left( \frac{1}{2N} \right) = \mu$$

- **Core Result:** The substitution rate  $k$  equals the neutral mutation rate  $\mu$  and is completely independent of population size  $N$ .
  - **The Molecular Clock Concept:** Because  $k = \mu$ , neutral mutations become fixed at a constant rate over time, allowing sequence divergence to act as a clock.
  - **Stochastic Nature:** Accumulation of substitutions is a stochastic process (contains inherent randomness).
  - **Factors Influencing Clock Rates:**
    - **Mutation Rate ( $\mu$ ):** Varies across organisms and viral types.
    - **Purifying Selection:** Removes harmful mutations, slowing the rate of substitution. Exons experience stronger purifying selection than introns, resulting in fewer observed substitutions.
-

## 8 Phylogenetic Analysis and Evolutionary Trees

- **Definition & Core Purpose:** Phylogenetics is the study of evolutionary histories among species or populations, represented via branching network graphs.
  - **Homology:** Fundamental requirement that sequence or phenotypic characters must be derived from a common ancestor.
  - **Tree Components & Terminology:**
    - **Nodes:** Represent internal common ancestors or speciation/split events.
    - **Branches (Edges):** Connect nodes; length represents time or inferred evolutionary change.
    - **Tips / Leaves / Taxa:** Observed extant or sampled entities.
    - **Clade:** A monophyletic group consisting of a common ancestor and all its descendants.
    - **Topology:** The branching shape and organization.
  - **Tree Types:**
    - **Cladograms:** Equal branch lengths showing common ancestry without indicating evolutionary time or change.
    - **Phylograms:** Branch lengths proportional to inferred evolutionary change (mutations per site).
    - **Time Trees:** Dated phylograms where branch lengths represent absolute dates (years or mya).
  - **Tree Building Methods:**
    - **Neighbor-Joining (NJ):** Fast distance-based method without explicit sequence change models.
    - **Maximum Likelihood (ML):** Calculates the most likely tree topology given an explicit statistical model of sequence change.
    - **Bayesian Inference:** Iteratively calculates posterior probabilities using explicit sequence change models.
  - **Tree Rooting Methods:** Outgroup rooting, midpoint rooting, and dated tips.
  - **Epidemic & Evolutionary Patterns Inferred from Topology:** Demonstrates viral dynamics and lineage turnover in pathogens (e.g., Influenza A, Dengue, SARS-CoV-2).
- 

## 9 Conservation Genetics and Molecular Forensics

- **The Extinction Vortex:** Small population size leads to increased drift and inbreeding, causing loss of genetic diversity and inbreeding depression, which further reduces population size and increases extinction risk.

- **Inbreeding Depression:** Reduced fitness in offspring from related matings, caused by unmasking deleterious recessives and loss of overdominance.
  - **Genetic Rescue:** Introduction of outside individuals into an inbred population to restore genetic diversity and eliminate inbreeding depression.
  - **Evolutionary Significant Units (ESUs):** Populations prioritized for conservation based on historical isolation, defined by reciprocal monophyly for mtDNA and significant nuclear allele frequency divergence.
  - **DNA Barcoding:** Standardized species identification using specific short gene sequences (~650 bp of mitochondrial COI in animals; *rbcL* and *matK* in plants). Based on the assumption that interspecific sequence divergence is significantly greater than intraspecific variation (“barcoding gap”).
- 

## 10 Integrative Real-World Applications & Quantitative Methods

- **Medical Genetics & Pathogen Tracking:** Mapping disease loci, quantifying pathogen substitution rates, and tracking transmission chains using time trees.
  - **Agriculture & Pest Management:** Detecting selection on pesticide resistance ( $F_{ST}$  outliers) and modeling crop pest adaptation.
  - **Conservation Management:** Incorporating  $N_e$ ,  $F_{ST}$ , ESUs, and genetic rescue to halt extinction vortices in threatened populations.
  - **Genomics & Computational Tools:** Synthesis of theoretical population models ( $N_e$ ,  $F_{ST}$ ,  $k = \mu$ ) with high-throughput sequencing, non-tree clustering (PCA, STRUCCURE), and phylogenetic software (MEGA, IQ-TREE, RAxML-NG).
- 

## 11 Population Genomics and Genome-Wide Selection Dynamics

- **Population Genomics Definition:** The expansion of population genetics from single/few loci to high-throughput, whole-genome sequencing across hundreds to thousands of individuals within a species.
- **Sequencing Technologies & Applications:**
  - **Illumina:** Primary short-read platform for population resequencing ( $5\times$ – $40\times$  depth), RNA-seq, ChIP-seq, and metagenomics.
  - **Pacific Biosciences (PacBio):** Long, accurate reads suitable for *de novo* reference assemblies.

- **Oxford Nanopore:** Generates ultra-long read lengths; ideal for rapid field sequencing and complex structural variant resolution.
- **Sanger Sequencing:** ABI-based lower-throughput method used for plasmid verification and targeted small-scale surveys.

- **Genome-Wide Summary Statistics:**

- **Nucleotide Diversity ( $\pi$ ):** Average number of nucleotide differences per site between any two randomly chosen sequences.
- **Segregating Sites ( $S$ ) & Watterson’s Estimator ( $\theta_W$ ):** Metric of diversity scaled by the number of mutating sites.
- **Allele Frequency Metrics:** Minor Allele Frequency (MAF) and Derived Allele Frequency (DAF).

- **Linkage Disequilibrium (LD) & Sweep Signatures:**

- **LD Decay:** Non-random association of alleles at distinct loci, broken down over time via recombination.
- **Selective Sweeps:** Positive selection drives a beneficial mutation to high frequency, sweeping adjacent linked neutral variants along with it.
- **Signatures:** Localized drop in nucleotide diversity ( $\pi$ ), marked increase in LD, excess of rare alleles, and high Extended Haplotype Homozygosity (EHH).

### 11.0.1 Genomic Patterns Across Modes of Selection

| Selection Mode              | Diversity ( $\pi$ , $\theta_W$ ) Effect          | Allele Frequency Spectrum                                      | Haplotype Structure                             |
|-----------------------------|--------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------|
| <b>Purifying (Negative)</b> | Reduced in functional regions (exons, promoters) | Excess of rare, low-frequency alleles (negative Tajima’s $D$ ) | Baseline decay                                  |
| <b>Adaptive (Positive)</b>  | Locally eliminated surrounding the swept allele  | Rapidly fixed/high DAF for beneficial allele                   | Long, conserved haplotypes (high EHH / $nS_L$ ) |
| <b>Balancing</b>            | Elevated surrounding target locus                | Excess of intermediate-frequency common alleles                | Divergent persistent haplotypes                 |

- **Polygenic Adaptation & Distribution of Fitness Effects (DFE):**

- **DFE Profile:** Most new mutations are neutral or slightly deleterious; strongly advantageous mutations ( $s > 0$ ) are extremely rare.

- **Polygenic Architecture:** Most complex traits adapt through subtle, simultaneous frequency shifts across many loci.
  - **Population Genomic Case Studies:**
    - **Drug Resistance Sweeps:** Pyrimethamine selection on *dhfr* in *P. falciparum*.
    - **Haplotype-Based Selection Tests:**  $nS_L$  statistic applied to *P. vivax* targets like the *AP2* transcription factor.
- 

## 12 Comparative Genomics, Evolutionary Rates, and Divergence

- **Diversity vs. Divergence:**
  - **Diversity:** Polymorphic variation present within a species population.
  - **Divergence:** Fixed sequence differences accumulated between isolated species over time.
- **Comparative Genomics Pipeline:** Assembly, annotation/homology search, synteny analysis, and multi-species reference alignments.
- **Determinants of Evolutionary Rates:**
  - **Functional Essentiality:** Critical genes evolve significantly slower due to intense purifying selection.
  - **Expression Level:** Highly expressed genes exhibit lower evolutionary rates.
  - **Protein Structural Constraints:** Core residues evolve slowly; surface-exposed residues evolve rapidly.
- **Quantifying Evolutionary Constraint:**
  - Conserved non-coding regions and coding exons show reduced divergence across multi-species alignments due to purifying selection.
- **Sequence Divergence Tests for Selection:**
  - **Synonymous Rate ( $dS$  or  $Ks$ ):** Amino acid-preserving substitutions; selectively neutral baseline.
  - **Nonsynonymous Rate ( $dN$  or  $Ka$ ):** Amino acid-altering substitutions; subject to functional selection.
  - $\frac{dN}{dS}$  **Ratio Interpretation:**
    - \*  $\frac{dN}{dS} < 1$ : Purifying selection.
    - \*  $\frac{dN}{dS} = 1$ : Neutral sequence divergence.
    - \*  $\frac{dN}{dS} > 1$ : Adaptive positive selection.
- **The McDonald-Kreitman (MK) Test:**

- Compares nonsynonymous-to-synonymous polymorphism within species ( $P_n/P_s$ ) against divergence between species ( $D_n/D_s$ ).
  - **Adaptive Divergence:**  $D_n/D_s > P_n/P_s$ .
  - **Balancing Selection:**  $P_n/P_s > D_n/D_s$ .
- 

## 13 Continuous Variation, Phenotypic Components, and Heritability

- **Continuous vs. Discontinuous Traits:** Continuous quantitative traits are determined by multiple genetic loci interacting with environmental factors.
- **Components of Phenotypic Variance ( $V_P$ ):**

$$V_P = V_G + V_E + V_{G \times E}$$

- **Components of Genetic Variance ( $V_G$ ):**

$$V_G = V_A + V_D + V_I$$

- **Additive Genetic Variance ( $V_A$ ):** Cumulative phenotypic effects of individual allele substitutions (primary basis for parent-offspring resemblance).
- **Dominance Genetic Variance ( $V_D$ ):** Results from allelic interactions at the same locus.
- **Epistasis Genetic Variance ( $V_I$ ):** Arises from inter-locus interactions.
- **Broad-Sense vs. Narrow-Sense Heritability:**
  - **Broad-Sense Heritability ( $H^2$ ):**  $H^2 = \frac{V_G}{V_P}$
  - **Narrow-Sense Heritability ( $h^2$ ):**  $h^2 = \frac{V_A}{V_P}$
- **Estimating  $h^2$  via Regressions:**
  - Mid-parent regression slope  $b$  gives  $h^2 = b$ .
  - Single-parent regression slope  $b$  gives  $h^2 = 2b$ .

- **The Breeder's Equation:**

$$R = h^2 S$$

- Selection Differential ( $S = M_S - M$ ) and Response to Selection ( $R = M' - M$ ).
-

## 14 Quantitative Trait Loci (QTL) Mapping and GWAS

- **Quantitative Trait Loci (QTL):** Specific chromosomal regions statistically associated with variation in a quantitative trait.
- **Bi-Parental Mapping Populations:**  $F_2$ , Backcross ( $BC$ ), Recombinant Inbred Lines ( $RILs$ ), and Near Isogenic Lines ( $NILs$ ).
- **QTL Mapping Methods:** Single Marker Analysis ( $ANOVA$ ), Interval Mapping ( $IM$ ), and Composite Interval Mapping ( $CIM$ ).
- **Genome-Wide Association Studies (GWAS):** Scans genome-wide SNPs across natural populations to detect marker-trait associations using historical recombination.
- **Linkage vs. Linkage Disequilibrium (LD):** Linkage refers to physical co-localization on a chromosome; LD represents statistical non-random association of alleles across a population.
- **Factors Governing LD Decay:** Recombination rate, population size ( $N_e$ ), mating system (selfing vs. outcrossing), natural selection, and population structure.
- **Methodological Controls & Missing Heritability:** Mixed linear models and PCA correct for population structure. Missing heritability describes the gap between pedigree  $h^2$  and GWAS SNP-based heritability ( $h^2_{\text{SNP}}$ ).

---

## 15 Genome Evolution and Structural Dynamics

- **Definition and Complexity (The C-Value Paradox):**
  - **Genome Size vs. Complexity:** Prokaryote genome size correlates linearly with complexity. Eukaryotes show no correlation between genome size and morphological complexity (**C-value paradox**).
  - **Diversity:** Ranges from small eukaryotic genomes to massive genomes (*Homo sapiens* 3.3 Gbp, *Ambystoma mexicanum* 32 Gbp, *Paris japonica* 149 Gbp).

### 15.0.1 Structural Organization: Prokaryotes vs. Eukaryotes

| Feature                     | Prokaryotic Genomes                    | Eukaryotic Genomes                         |
|-----------------------------|----------------------------------------|--------------------------------------------|
| <b>Chromosome Structure</b> | Typically single circular chromosome   | Multiple linear chromosomes                |
| <b>DNA Packaging</b>        | Condensed in nucleoid via supercoiling | Condensed in nucleus via histone complexes |

| Feature                          | Prokaryotic Genomes                        | Eukaryotic Genomes                                    |
|----------------------------------|--------------------------------------------|-------------------------------------------------------|
| <b>Ploidy &amp; Gene Copy</b>    | Haploid (single gene copy)                 | Mostly diploid (two copies per gene)                  |
| <b>Expression Coupling</b>       | Simultaneous transcription and translation | Spatially separated (nucleus vs. cytoplasm)           |
| <b>Extrachromosomal Elements</b> | Common nonessential plasmids               | Plasmids rare (mostly organellar genomes)             |
| <b>Genome Compactness</b>        | Highly compact; minimal noncoding DNA      | Extensive non-coding introns, regulatory regions, TEs |

- **Prokaryotic Genome Evolution and Lifestyle:**

- Free-living prokaryotes maintain larger genomes (e.g., *Sorangium cellulosum* 13 Mb); obligate symbionts undergo gene decay/streamlining (e.g., *Carsonella ruddii* 0.16 Mb).
- **Horizontal Gene Transfer (HGT):** Transformation, Transduction, and Conjugation.

- **Organellar Origins:**

- Endosymbiotic theory: Mitochondria ( $\alpha$ -proteobacteria) and chloroplasts (cyanobacteria).

- **Eukaryotic Repetitive Elements and Transposable Elements (TEs):**

- **Class I (Retrotransposons):** “Copy-and-paste” via RNA intermediate (LINEs, SINEs, LTRs).
- **Class II (DNA Transposons):** “Cut-and-paste” mechanism.

- **Spliceosomal Introns & Alternative Splicing:**

- Alternative splicing expands protein diversity (e.g., *Drosophila Dscam1* generating thousands of isoforms).
- Spliceosomal introns originated from ancestral bacterial Group II self-splicing introns.

- **Linear Chromosomes, Sex Chromosomes, & Polyploidy:**

- Telomerase resolves end-replication problems on linear chromosomes.
- Suppression of recombination around sex-determining loci leads to heteromorphic sex chromosomes (X/Y, Z/W).
- **Whole Genome Duplication (WGD):** Autopolyploidy and Allopolyploidy (e.g., *Triticum aestivum*) drive instant speciation and gene duplication.